

Document 7: Routing Compiler Scoring Function Specification

7.1 Primary Routing Valuation

Each active pedal calculates a primary routing score scaled from 0 to 100 based on the average horizontal vertex position of its painted pixels.

Let P_{total} represent the total number of non-zero cells within the pedal's assigned row boundaries, and let X_k represent the horizontal grid index (0 to 127) of the k -th painted cell. The primary value is calculated as:

$$Value_{primary} = \lfloor ((\sum X_k) / P_{total}) \times (100 / 127) \rfloor$$

If $P_{total} = 0$ (empty canvas space), $Value_{primary} = 0$.

7.2 Priority Sorting Rules

Pedals are sorted sequentially in descending order according to their primary valuation. The pedal with the greatest derived value receives the highest priority and is placed first in the serial audio chain (closest to the input source).

7.3 Multi-Tiered Tie-Breaker Resolution

If two or more pedals derive the exact same primary valuation, the collision is resolved deterministically by falling back to secondary and tertiary visual parameters in the following sequence:

- **Tie-Breaker 1 (Pixel Density):** The pedal with the higher ratio of painted cells to total assigned cells takes priority.
- **Tie-Breaker 2 (Color Weight Accumulation):** The pedal with the higher net absolute sum of color weights (+2 for White, +1 for Red, 0 for Green, -1 for Blue, -2 for Black) takes priority.
- **Tie-Breaker 3 (Physical Index Default):** If an absolute tie persists, priority falls back to the lowest physical slot index ordering (Slot 0 through Slot 5).